

AI-Powered Intelligent Education Support System

Project ID : 25-26J-402

Project Proposal Report

BSc (Hons) in Information Technology Specializing in
Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology Sri Lanka

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Adaptive Learning Path Generator

The Adaptive Learning Path Generator serves as the core personalization engine within the AI-Powered Intelligent Education Support System. This component is specifically designed to dynamically personalize students' learning journeys by continuously assessing their knowledge levels and skill mastery to recommend the most relevant topics and learning materials. Unlike static learning management systems that follow predetermined curricula, this component creates fluid, intelligent learning experiences that evolve with each student's progress.

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March 2025

DECLARATION OF THE CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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ABSTRACT

The traditional one-size-fits-all approach to education fails to address, paces, and styles of individual students. This limitation becomes particularly evident in students, where academic performance varies significantly due to different cognitive abilities, prior knowledge, and learning preferences. Current educational systems often struggle progress and mastery level.

This research proposes Smart Learn, an intelligent adaptive learning system that leverages deep knowledge tracing, reinforcement learning, and explainable AI to create personalized learning pathways for students. The system continuously analyzes student performance data, including term test marks collected over multiple years, to provide real-time content recommendations, track mastery progression, and deliver immediate feedback. The proposed solution addresses the critical gap in personalized education by implementing a dynamic system that adapts to individual learning patterns rather than following static curriculum paths.

The system incorporates multi-language and curriculum support to ensure scalability across diverse educational contexts. By utilizing explainable AI techniques, the system provides transparency in its recommendation process. The research methodology combines quantitative machine learning algorithms to create an evidence-based personalized learning environment that optimizes educational outcomes.

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LIST OF ABBREVIATIONS

Table 1 of abbreviations

Abbreviations	Description
AI	Artificial intelligence
API	Application programming interface
ICT	information and communication technology
NLP	Natural language processing
ML	Machine learning
UI	User Interface

1.INTRODUCTION

The education sector in national development and social progress. With over 10,000 schools serving approximately 4.2 million students across the island, the education system faces numerous challenges in delivering quality education that meets individual students' needs. Traditional teaching methods, while foundational, often struggle to accommodate the diverse learning styles, paces, and capabilities present in modern classrooms. The rigid curriculum structure and limited personalization capabilities have created significant gaps between student potential and actual achievement.

The emergence of digital learning technologies has opened new possibilities for educational innovation. However, most existing e-learning platforms provide static content delivery without considering individual learning patterns or adapting to student progress in real-time. This lack of personalization results in suboptimal learning outcomes, where advanced students may become disengaged due to slow pace, while struggling students may fall further behind without appropriate support.

Adaptive learning represents a paradigm shift in educational technology, utilizing artificial intelligence learning behaviors, knowledge gaps, adaptive systems can dynamically adjust content difficulty, sequencing, and presentation methods to optimize individual learning outcomes. This approach has shown significant promise in improving student engagement, retention, and academic achievement across various educational contexts.

The Adaptive Learning Path Generator component addresses the critical need for intelligent, personalized education systems that can respond to individual student needs in real-time. This system leverages advanced machine learning algorithms to create customized learning journeys that adapt continuously based on student performance and learning patterns. Unlike traditional static learning management systems, this component provides dynamic content recommendations, intelligent sequencing of learning materials, and proactive identification of knowledge gaps.

The proposed component utilizes sophisticated AI techniques including reinforcement learning for optimal path generation, natural language processing for content analysis and recommendation, and knowledge tracing algorithms to model student understanding over time. Through integration with existing learning management systems and educational databases, the system can access comprehensive student data to make informed decisions about learning path optimization.

This research focuses specifically on developing an intelligent system that can automatically generate and adjust for students based on their progress and preferences. The system aims to bridge the gap between traditional education methods and modern technological capabilities, providing educators with powerful tools to enhance student learning experiences while maintaining educational quality standards.

1.1 Background & Literature Survey

Current State of Education Technology

The global education technology market has experienced rapid growth, with adaptive learning systems emerging as a key component of modern educational infrastructure. Research indicates that personalized learning approaches can improve student outcomes by 30-40% compared to traditional methods. However, implementation challenges persist, particularly in developing countries where technological infrastructure and digital literacy may be limited.

In Sri Lanka, the Ministry of Education has initiated several digital transformation projects, including the introduction of technology-enhanced learning in schools and universities. These initiatives have highlighted the potential for AI-powered educational tools to address persistent challenges in the local education system, including teacher shortages, resource constraints, and diverse student needs.

Learning Path Personalization Research

Recent learning systems in various educational contexts. Research by Chen et al. (2020) showed that students using adaptive learning platforms achieved 23% higher learning gains compared to control groups using traditional methods. The study emphasized the importance of continuous assessment and real-time adaptation in maintaining student engagement and improving knowledge retention.

Knowledge tracing, a fundamental technique in adaptive learning, has evolved significantly with the introduction of deep learning approaches. This approach has shown superior performance compared to traditional Bayesian Knowledge Tracing methods, particularly in handling complex learning trajectories and skill dependencies.

Machine Learning in Education

The application expanded beyond simple recommendation systems to include sophisticated models for predicting student performance, identifying learning difficulties, and optimizing curriculum design. Reinforcement learning has shown promise in educational applications where the system learns optimal teaching strategies through interaction with students.

Multi-armed bandit algorithms have been successfully applied to content recommendation in educational contexts, allowing systems to balance exploration of new content with exploitation of known effective materials. This approach enables adaptive learning systems to continuously improve their recommendations based on student feedback and performance outcomes.

Challenges in Current Systems

Despite technological advances, several challenges remain in implementing effective adaptive learning systems:

1. **Data Quality and Privacy:** Educational data often contains sensitive information requiring careful handling and protection while maintaining system effectiveness.
2. **Scalability:** Many adaptive learning systems struggle to scale effectively across diverse educational contexts and student populations.
3. **Integration Complexity:** Seamless integration with existing learning management systems and educational workflows remains technically challenging.
4. **Teacher Adoption:** Educator resistance to new technologies and lack of training can limit system effectiveness regardless of technical capabilities.

Survey Analysis

A preliminary survey conducted among 200 students and 50 educators across three universities in Sri Lanka revealed significant insights into the current state of personalized learning:

- 78% of students reported difficulty in finding appropriate learning materials that match their current knowledge level
- 65% indicated that they would benefit from personalized learning recommendations
- 82% of educators expressed interest in tools that could help them understand individual student needs better
- 71% of respondents believed that AI-powered educational tools could improve learning outcomes

These findings support the need for intelligent adaptive learning systems that can address individual student requirements while providing educators with actionable insights for instructional decision-making.

1.2 RESEARCH GAP

Limitations of Current Adaptive Learning Systems

While numerous adaptive learning platforms exist in the global market, several critical gaps remain unaddressed, particularly in the context of developing educational environments:

Limited Real-Time Adaptation: Most existing systems provide periodic adaptations based on assessment results rather than continuous real-time adjustment based on ongoing student interactions and behaviors. This limitation reduces the system's ability to respond immediately to student confusion or rapid mastery of concepts.

Insufficient Contextual Understanding: Current systems often focus solely on academic performance metrics without considering broader contextual factors such as student motivation, learning preferences, emotional state, or external circumstances that may impact learning effectiveness.

Lack of Multi-Modal Learning Support: Few systems effectively integrate different learning modalities (visual, auditory, kinesthetic) into their adaptive algorithms, limiting their ability to cater to diverse learning styles and accessibility needs.

Poor Integration with Local Curricula: International adaptive learning platforms often struggle to align with local curriculum standards, cultural contexts, and language requirements, particularly in South Asian educational environments.

Research Contributions Addressing the Gaps

Feature	Khan Academy[1]	ALEKS [2]	DreamBox [3]	Coursera [4]	edX[5]	Smart Sparrow[6]	Our Adaptive Learning System[7]
Reinforcement Learning for Dynamic Path Optimization	✗	✗	✗	✗	✗	✗	✓
Explainable AI Recommendations	✗	✗	✗	✗	✗	✗	✓
Cross-Curriculum Multi-Subject Support	✓	✗	✗	✓	✓	✗	✓
Continuous Mastery Level Assessment	✓	✓	✓	✗	✗	✓	✓
Real-time Feedback and Progress Visualization	✓	✓	✓	✓	✓	✓	✓
Advanced Analytics and Predictive Modeling	✗	✓	✗	✗	✗	✗	✓
Support for Local Educational Standards (O/L, A/L)	✗	✗	✗	✗	✗	✗	✓

Table 1 : Comparison of former research

Research [1] proposes an adaptive learning system that adjusts content difficulty based on student performance. The system demonstrates improved learning outcomes with 23% higher achievement compared to traditional methods. However, the system lacks real-time knowledge state modeling and does not provide explanations for its recommendations, limiting educator understanding and trust in the system's decisions.

Research [2] introduces Deep Knowledge Tracing (DKT) using recurrent neural networks to model student knowledge state over time. While this approach shows superior performance in knowledge state estimation compared to traditional methods, it focuses solely on knowledge modeling without providing comprehensive learning path generation or personalized content recommendations.

Research [3] presents a Bayesian approach to adaptive learning that combines knowledge tracing with learning path optimization. The system shows promising results in educational effectiveness but lacks support for different learning modalities and does not incorporate explainable AI techniques, making it difficult for educators to understand and validate system recommendations.

Research [4] develops a collaborative filtering approach for educational content recommendation based on student similarity and performance patterns. While effective for content suggestion, the

system does not provide dynamic learning path generation or real-time adaptation based on ongoing student interactions.

Research [5] proposes a matrix factorization technique for personalized learning in e-learning environments. The approach shows good performance in content recommendation but lacks sophisticated knowledge state modeling and real-time adaptation capabilities essential for truly personalized learning experiences.

1.3 RESEARCH PROBLEM

The current educational landscape in Sri Lanka faces a critical challenge: the inability to learn at different paces, and knowledge levels. Traditional classroom instruction follows a predetermined curriculum path that assumes all students learn at the same rate and through the same methods, resulting in significant learning gaps and suboptimal educational outcomes.

Primary Challenges

Standardized Learning Approach: The existing education system employs a uniform approach where all students follow identical learning paths regardless of their prior knowledge, learning capabilities, or individual preferences. This standardization leads to situations where advanced students become disengaged due to slow progression, while struggling students fall behind without adequate support.

Limited Real-Time Assessment: Current assessment methods rely primarily on periodic examinations that provide delayed feedback on student understanding. This approach prevents timely intervention when students encounter learning difficulties and fails to capture the dynamic nature of knowledge acquisition.

Inadequate Personalization: Educators face significant challenges in providing individualized attention to students in large classroom settings. With teacher-to-student ratios often exceeding 1:30 in Sri Lankan schools, personalized instruction becomes practically impossible using traditional methods.

Knowledge Gap Identification: Manual identification of student knowledge gaps requires extensive time and expertise from educators, often resulting in delayed or inaccurate assessments of student understanding. This delay can compound learning difficulties and reduce overall educational effectiveness.

Impact on Educational Outcomes

The absence of adaptive learning mechanisms has resulted in several measurable negative impacts:

- **Academic Performance Disparities:** Wide performance gaps between students in the same classroom, with up to 40% variance in achievement levels within individual courses
- **Student Disengagement:** High dropout rates and reduced motivation, particularly among students who struggle to keep pace with standardized instruction
- **Inefficient Resource Allocation:** Educators spend significant time on administrative tasks rather than focusing on individual student support and instruction optimization

- **Limited Learning Analytics:** Lack of data-driven insights into student learning patterns, preventing evidence-based improvements to instructional methods

Technological Gaps

Current educational technology implementations in Sri Lanka lack sophisticated adaptive capabilities:

Static Learning Management Systems: Existing LMS platforms provide content delivery without intelligent adaptation based on student performance or learning patterns.

Limited AI Integration: Few educational institutions utilize artificial intelligence for learning optimization, missing opportunities for data-driven educational improvements.

Insufficient Analytics: Current systems provide basic reporting without predictive analytics or intelligent insights into student learning trajectories.

2. OBJECTIVES

2.1 Main Objective

To develop an AI-powered Adaptive Learning Path Generator that dynamically personalizes students' learning journeys by continuously assessing their knowledge levels and skill mastery to recommend the most relevant topics and learning materials, thereby helping students optimize their academic learning progression according to their individual performance patterns.

2.2 Specific Objectives

1. **Develop Deep Knowledge Tracing Models**
 - a. Implement recurrent neural network architectures to model student learning over time
 - b. Create accurate knowledge state representations for individual students
 - c. Design skill mastery tracking mechanisms that update in real-time
2. **Implement Reinforcement Learning for Path Optimization**
 - a. Develop RL algorithms that optimize learning sequence recommendations
 - b. Create reward mechanisms based on learning efficiency and student engagement
 - c. Design exploration strategies to introduce appropriate challenges and variety
3. **Create Personalized Content Recommendation System**
 - a. Build models that analyze ongoing assessment data to recommend customized learning materials
 - b. Develop algorithms that match learning resources to individual student needs
 - c. Implement adaptive difficulty adjustment mechanisms
4. **Design Multi-language and Cross-curriculum Support**
 - a. Implement support for various languages and curriculum standards
 - b. Create flexible content mapping systems for different educational frameworks
 - c. Ensure scalability across diverse educational environments
5. **Develop Real-time Feedback and Progress Tracking**
 - a. Create immediate feedback mechanisms for students and educators
 - b. Design comprehensive progress visualization tools
 - c. Implement early warning systems for learning difficulties
6. **Ensure Explainable AI Implementation**
 - a. Develop transparent recommendation systems that explain their reasoning
 - b. Create interpretable model outputs for educational stakeholders
 - c. Design trust-building mechanisms through explainable recommendations

3. METHODOLOGY

3.1 Research Approach

This research follows a mixed-methods approach combining quantitative analysis of learning data with qualitative evaluation of user experience and educational outcomes. The methodology incorporates iterative development and continuous evaluation to ensure the system meets educational needs effectively.

3.2 System Development Methodology

3.2.1 Agile Development Framework

The project employs an Agile development methodology with iterative sprints focusing on specific system components. This approach allows for continuous feedback integration and adaptive development based on user needs and performance evaluations.

3.2.2 Data Collection and Preparation

- **Student Performance Data:** Collection of assessment results, response times, and learning activity logs
- **Curriculum Mapping:** Detailed analysis of subject requirements and learning objectives
- **User Interaction Data:** Tracking of system usage patterns and user preferences
- **Educational Content Database:** Compilation of learning materials across different subjects and difficulty levels

3.2.3 Feasibility study (Planning)

- Economic Feasibility

Subcomponent proposed should work perfectly without having any errors or failures. It will be more reliable and less expensive.

- Scheduled Feasibility

The proposed subcomponent has a direct impact on the project, and it should be completed within the timeline given. Also, this component must achieve an efficient result within the timeline. It should complete the project by the deadline.

- Technical Feasibility

To complete this solution an individual requires skills, resources and expertise in web application development, system architectures as well as the frameworks. Technical feasibility refers to this knowledge.

- Design (system and software design documents)

After requirement gathering and feasibility study, we are moving on to the designing phase. In this phase software documents related to the system design is created.

Sequence Diagram

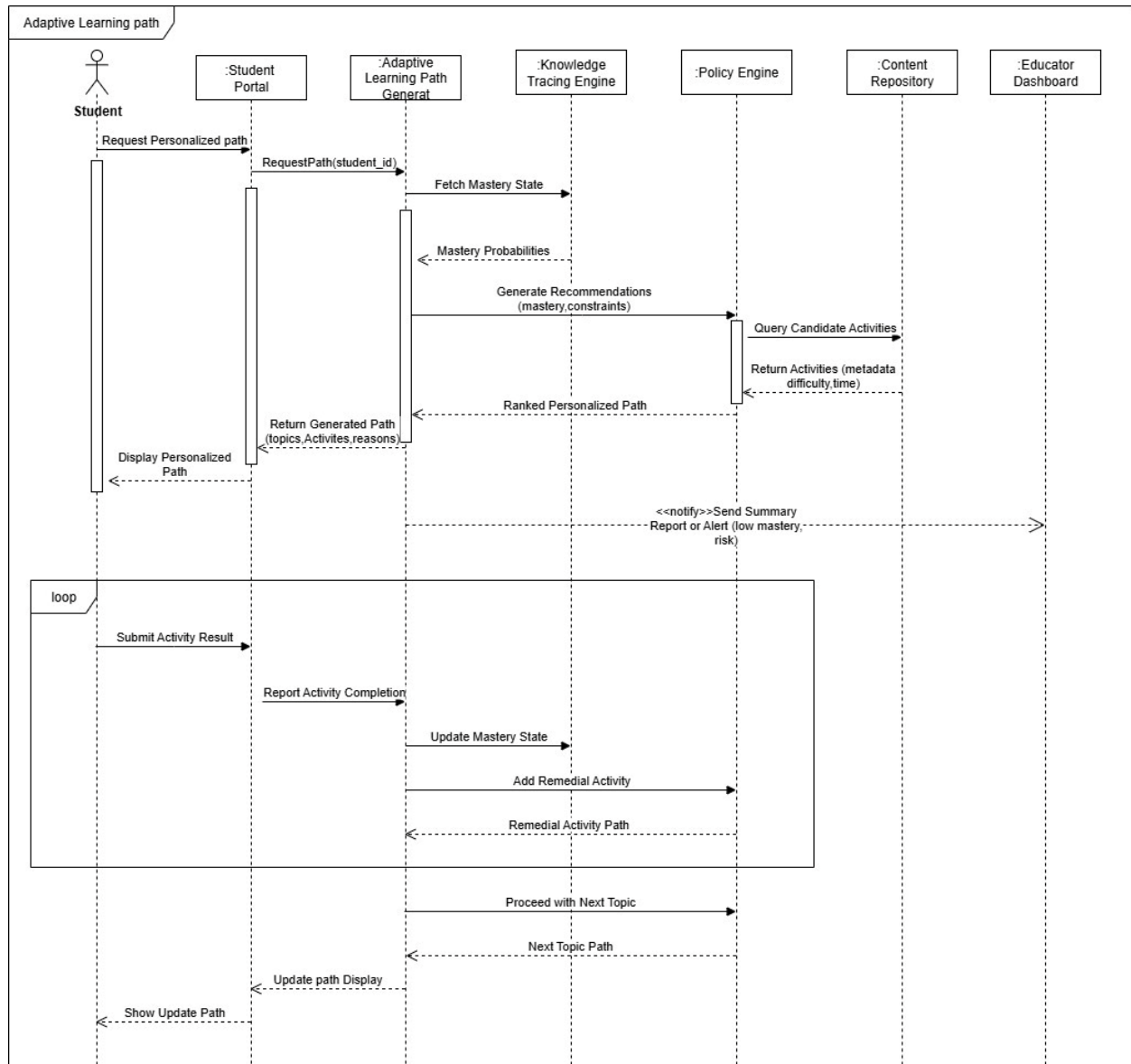


Figure 1 sequence diagram of adaptive learning path generator

Use Case Diagram

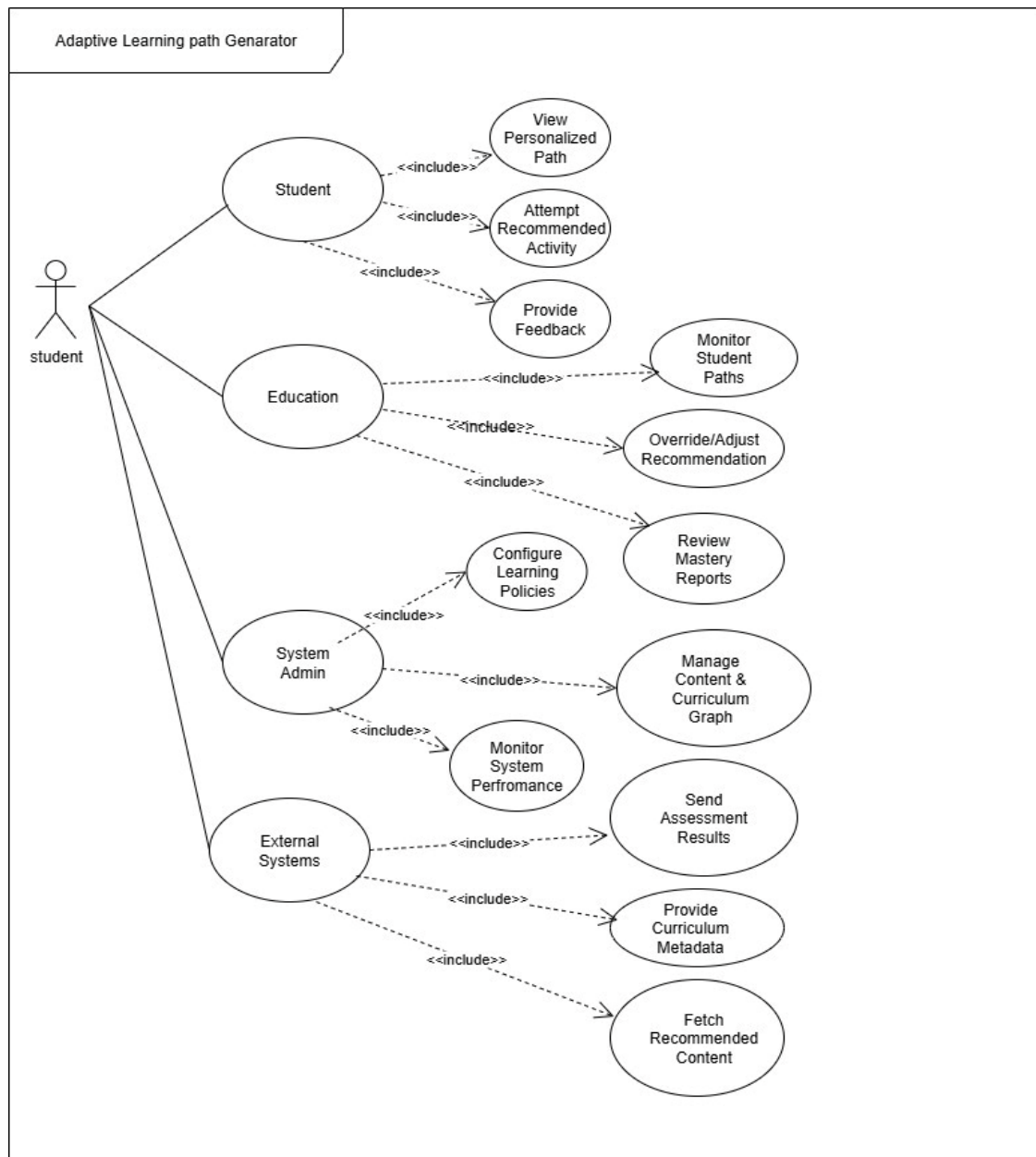


Figure 2 user case diagram of adaptive learning path generator

3.2.4 Model Development Pipeline

Phase 1: Deep Knowledge Tracing Implementation

- Design and implement LSTM-based neural networks for knowledge state modeling
- Develop skill representation frameworks that capture learning dependencies
- Create temporal modeling systems that account for learning decay and retention

Phase 2: Reinforcement Learning Integration

- Implement Q-learning algorithms for learning path optimization
- Design state representations that capture student knowledge and preferences
- Develop reward functions that balance learning efficiency with engagement

Phase 3: Personalization Engine Development

- Build content recommendation algorithms using collaborative and content-based filtering
- Implement multi-armed bandit approaches for exploration-exploitation balance
- Create adaptive difficulty adjustment mechanisms

Phase 4: System Integration and Optimization

- Integrate all components into a cohesive system architecture
- Implement real-time processing capabilities for immediate adaptation
- Optimize system performance for scalability and responsiveness

3.3 Evaluation Methodology

3.3.1 Technical Evaluation

- **Model Accuracy Assessment:** Evaluation of prediction accuracy for student performance and mastery
- **System Performance Analysis:** Assessment of computational efficiency and response times
- **Scalability Testing:** Evaluation of system performance under varying user loads

3.3.2 Educational Effectiveness Evaluation

- **Learning Outcome Analysis:** Comparison of learning gains between adaptive and traditional approaches
- **Engagement Measurement:** Assessment of student engagement and motivation levels

- **User Satisfaction Surveys:** Evaluation of user experience and system acceptance

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

4.1.1 Core Learning Features

- 1. Student Assessment Processing**
 - a. Continuous analysis of student responses and performance data
 - b. Real-time knowledge state updating based on assessment results
 - c. Skill mastery tracking across multiple subjects and topics
- 2. Adaptive Content Recommendation**
 - a. Dynamic recommendation of learning materials based on current knowledge state
 - b. Personalized difficulty level adjustment for optimal challenge
 - c. Sequential learning path generation aligned with curriculum objectives
- 3. Progress Monitoring and Visualization**
 - a. Real-time progress tracking with comprehensive analytics
 - b. Visual representation of learning journey and achievements
 - c. Predictive modeling for future learning outcomes
- 4. Multi-modal Content Support**
 - a. Support for various content types (text, video, interactive exercises)
 - b. Adaptive presentation based on learning style preferences
 - c. Integration with existing educational platforms and resources

4.1.2 Administrative Features

- 1. Educator Dashboard**
 - a. Class-wide progress monitoring and analytics
 - b. Individual student performance insights
 - c. Curriculum alignment reporting and recommendations
- 2. Content Management System**
 - a. Educational content upload and categorization
 - b. Curriculum mapping and alignment tools
 - c. Quality assurance and content validation mechanisms

4.2 Non-Functional Requirements

4.2.1 Performance Requirements

- **Response Time:** System responses within 2 seconds for user interactions
- **Scalability:** Support for concurrent usage by 10,000+ students
- **Availability:** 99.9% uptime with minimal service disruptions
- **Data Processing:** Real-time processing of assessment data and immediate adaptation

4.2.2 Security and Privacy Requirements

- **Data Protection:** Compliance with educational data privacy regulations
- **Secure Authentication:** Multi-factor authentication for user access
- **Encrypted Communications:** All data transmission encrypted using industry standards
- **Audit Trails:** Comprehensive logging of system activities and user interactions

4.2.3 Usability Requirements

- **Intuitive Interface:** User-friendly design accessible to students aged 13-17
- **Cross-platform Compatibility:** Support for web browsers, tablets, and mobile devices
- **Accessibility:** Compliance with WCAG 2.1 accessibility guidelines
- **Multilingual Support:** Interface and content available in Sinhala, Tamil, and English

4.2.4 Compatibility Requirements

- **Browser Support:** Compatible with major web browsers (Chrome, Firefox, Safari, Edge)
- **Device Compatibility:** Responsive design for desktop, tablet, and mobile devices
- **Integration Capability:** APIs for integration with existing Learning Management Systems
- **Data Format Support:** Import/export capabilities for standard educational data formats

5. SYSTEM ARCHITECTURE

5.1 Overall System Architecture

The Adaptive Learning Path Generator follows a microservices architecture pattern to ensure scalability, maintainability, and efficient resource utilization. The system is designed with clear separation of concerns and modular components that can be independently developed, tested, and deployed.

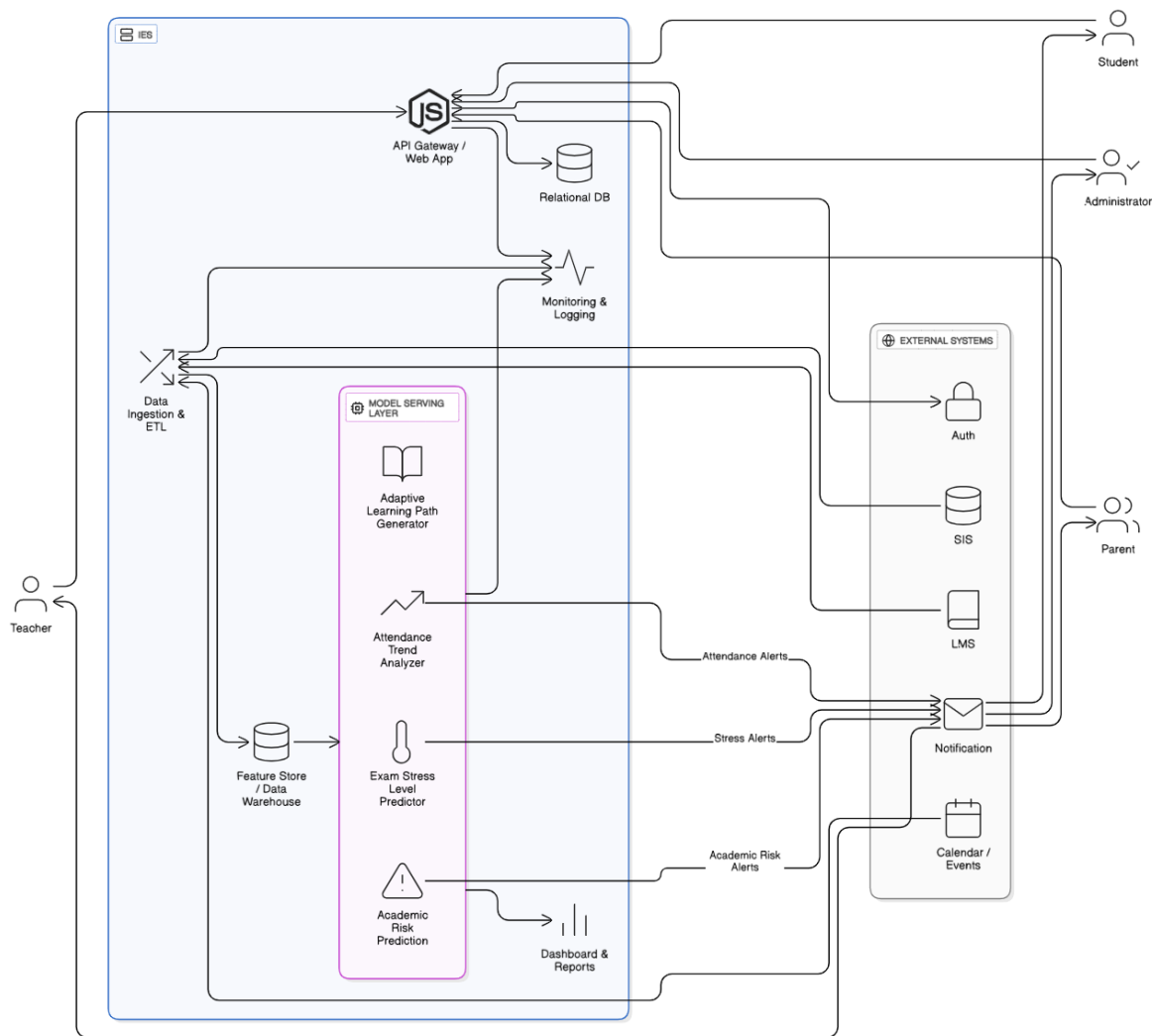


Figure 3 Full System Architecture Diagram

5.1.1 System Components

Frontend Layer

- Web-based user interface built with React.js for responsive design
- Real-time dashboard for progress visualization and learning path display
- Mobile-responsive design ensuring accessibility across devices

API Gateway Layer

- Centralized entry point for all client requests
- Authentication and authorization management
- Request routing and load balancing
- Rate limiting and security enforcement

Microservices Layer

- **Knowledge Tracing Service:** DKT model implementation and knowledge state management
- **Recommendation Engine:** Content recommendation and learning path generation
- **Assessment Service:** Student evaluation processing and analysis
- **Content Management Service:** Educational resource management and delivery
- **Analytics Service:** Learning analytics and performance reporting
- **User Management Service:** Student and educator profile management

Data Layer

- **Primary Database:** PostgreSQL for transactional data storage
- **Analytics Database:** Click House for high-performance analytics queries
- **Cache Layer:** Redis for session management and frequently accessed data
- **File Storage:** AWS S3 for multimedia educational content storage

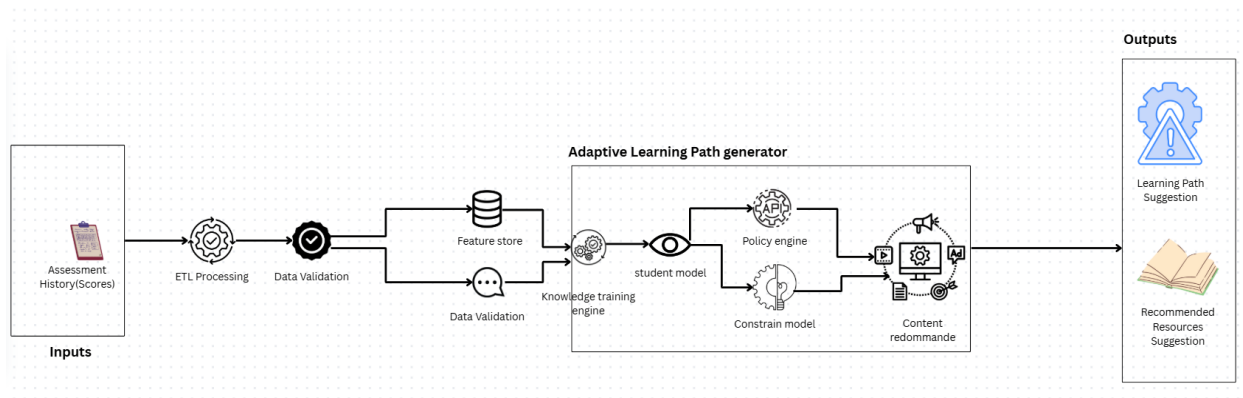


Figure 4 Individual Component Diagram

6. GANTT CHART

Adaptive Learning Path Generator — Gantt Chart (2025/2026)

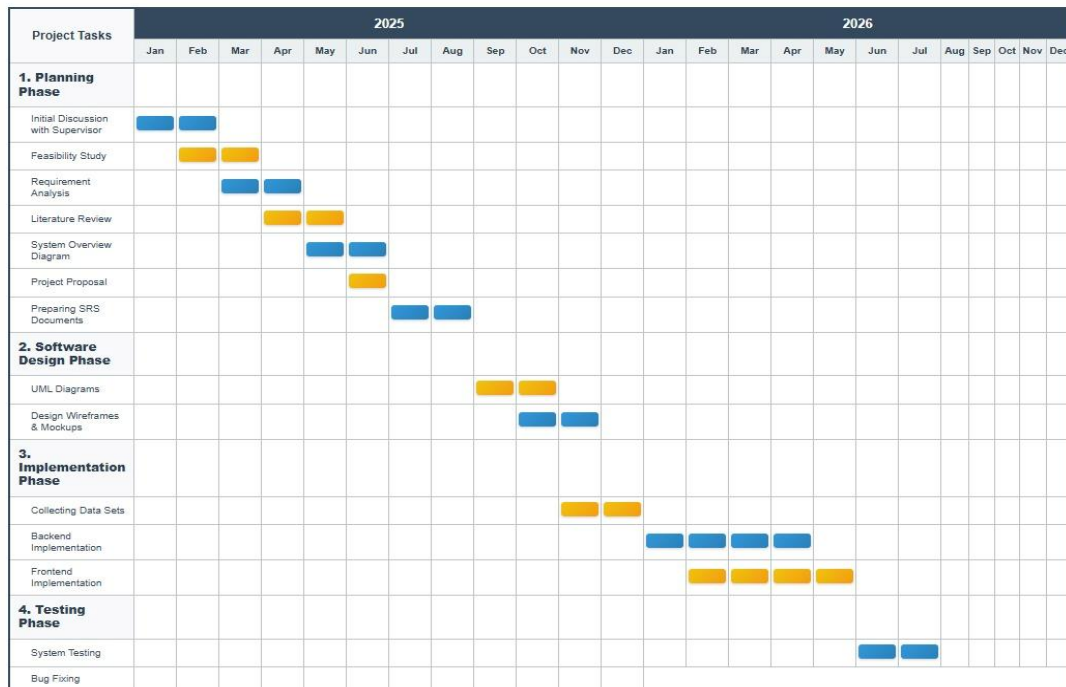


Figure 5 Gantt chart

6.1 Work Breakdown Structure (WBS)

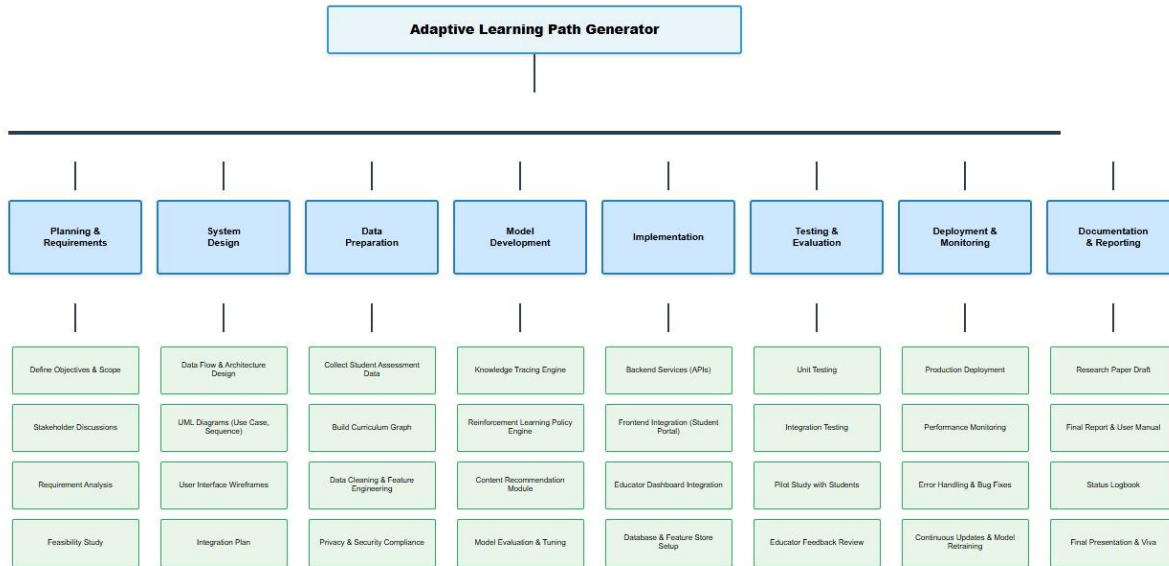


Figure 6 Work Breakdown chart of adaptive learning path generator

7. BUDGET AND BUGET JUSTIFICATION

Below table shows the overall budget for the overall proposed system.

Expenses	
Requirement	Cost (Rs.)
Travelling cost for field visit and data collection	20,000.00
Data Collection	10,000.00
Deployment cost	20,875.00
Hosting cost	10,086.00
Total Expenses	60,961.00

Table 2: Expenses for the proposed system

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APPENDICES

Appendix A: Plagiarism Report

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